# NexHire AI – Entity Relationship Diagram

Smart Employee Recruitment & AI Hiring System · MySQL relational schema · Crow's foot notation

**LEGEND**

| | | | |
|---|---|---|---|
| Exactly one (mandatory) | | Zero or one (optional) | |
| One or many (mandatory) | | Zero or many (optional) | |
| PK | Primary key (bold, underlined) | FK | Foreign key (italic) |
| UK | Unique key | PK,FK | Composite key part |

## Users & Access

**CALENDAR_CONNECTION**

| | | |
|---|---|---|
| PK | **connection_id** | INT |
| FK,UK | *user_id* | INT |
| | google_account | VARCHAR(150) |
| | access_token_enc | TEXT |
| | refresh_token_enc | TEXT |
| | token_expiry | DATETIME |

**PASSWORD_RESET**

| | | |
|---|---|---|
| PK | **reset_id** | INT |
| FK | *user_id* | INT |
| UK | token_hash | VARCHAR(255) |
| | expires_at | DATETIME |
| | used_at | DATETIME |

**USER**

| | | |
|---|---|---|
| PK | **user_id** | INT |
| | full_name | VARCHAR(100) |
| UK | email | VARCHAR(150) |
| UK | cnic | VARCHAR(15) |
| | password_hash | VARCHAR(255) |
| | role | ENUM(HR_ADMIN, INTERVIEWER, CANDIDATE) |
| | status | ENUM(ACTIVE, DEACTIVATED) |
| | created_at | DATETIME |
| | last_login_at | DATETIME |

**CANDIDATE_PROFILE**

| | | |
|---|---|---|
| PK | **profile_id** | INT |
| FK,UK | *user_id* | INT |
| UK | cnic | VARCHAR(15) |
| | phone | VARCHAR(20) |
| | address | VARCHAR(255) |
| | city | VARCHAR(60) |
| | highest_qualification | VARCHAR(100) |
| | updated_at | DATETIME |

## Offers, Referrals & Candidate Support

**FAQ_ENTRY**

| | | |
|---|---|---|
| PK | **faq_id** | INT |
| FK | *updated_by* | INT |
| | question | TEXT |
| | answer | TEXT |
| | category | VARCHAR(60) |
| | updated_at | DATETIME |

**OFFER**

| | | |
|---|---|---|
| PK | **offer_id** | INT |
| FK,UK | *application_id* | INT |
| FK | *approved_by* | INT |
| | salary_band | VARCHAR(50) |
| | start_date | DATE |
| | role_terms | TEXT |
| | letter_body | TEXT |
| | status | ENUM(DRAFT, APPROVED, SENT) |
| | sent_at | DATETIME |

**CHAT_SESSION**

| | | |
|---|---|---|
| PK | **session_id** | INT |
| FK | *candidate_id* | INT |
| | started_at | DATETIME |

**CHAT_MESSAGE**

| | | |
|---|---|---|
| PK | **message_id** | INT |
| FK | *session_id* | INT |
| | sender | ENUM(CANDIDATE, BOT) |
| | message_text | TEXT |
| | sent_at | DATETIME |

**REFERRAL**

| | | |
|---|---|---|
| PK | **referral_id** | INT |
| FK,UK | *application_id* | INT |
| FK | *marked_by* | INT |
| | employee_name | VARCHAR(100) |
| | employee_department | VARCHAR(100) |
| | employee_position | VARCHAR(100) |
| | decision | ENUM(PENDING, ACCEPTED, REJECTED) |
| | decided_at | DATETIME |

## Notifications, Integrations & Audit

**AUDIT_LOG**

| | | |
|---|---|---|
| PK | **log_id** | INT |
| FK | *user_id* | INT |
| | entity_type | VARCHAR(50) |
| | entity_id | INT |
| | action | VARCHAR(80) |
| | details | TEXT |
| | created_at | DATETIME |

**EMAIL_LOG**

| | | |
|---|---|---|
| PK | **email_id** | INT |
| FK | *recipient_id* | INT |
| FK | *application_id* | INT |
| | email_type | ENUM(CONFIRMATION, SHORTLIST, REJECTION, INTERVIEW, OFFER, PASSWORD_RESET) |
| | subject | VARCHAR(200) |
| | status | ENUM(QUEUED, SENT, FAILED) |
| | retry_count | INT |
| | sent_at | DATETIME |

## Applications & AI Screening

**STAGE_HISTORY**

| | | |
|---|---|---|
| PK | **history_id** | INT |
| FK | *application_id* | INT |
| FK | *changed_by* | INT |
| | from_stage | VARCHAR(20) |
| | to_stage | VARCHAR(20) |
| | changed_at | DATETIME |

**APPLICATION**

| | | |
|---|---|---|
| PK | **application_id** | INT |
| FK | *vacancy_id* | INT |
| FK | *candidate_id* | INT |
| | portfolio_link | VARCHAR(255) |
| | stage | ENUM(APPLIED, SHORTLISTED, INTERVIEW, OFFER, HIRED, REJECTED) |
| | submitted_at | DATETIME |
| | updated_at | DATETIME |

**DUPLICATE_FLAG**

| | | |
|---|---|---|
| PK | **flag_id** | INT |
| FK | *application_id* | INT |
| FK | *matched_application_id* | INT |
| | match_reason | VARCHAR(100) |
| | similarity | DECIMAL(5,2) |
| | status | ENUM(OPEN, MERGED, DISCARDED, DISMISSED) |
| FK | *resolved_by* | INT |
| | resolved_at | DATETIME |

**APPLICATION_ANSWER**

| | | |
|---|---|---|
| PK | **answer_id** | INT |
| FK | *application_id* | INT |
| FK | *field_id* | INT |
| | answer_value | TEXT |

**AI_SCORE**

| | | |
|---|---|---|
| PK | **score_id** | INT |
| FK,UK | *application_id* | INT |
| | fulfilment_pct | DECIMAL(5,2) |
| | rank_no | INT |
| | match_tag | ENUM(STRONG, POSSIBLE, NOT_A_FIT) |
| | predicted_score | DECIMAL(5,2) |
| | eval_status | ENUM(PENDING, EVALUATED, FAILED) |
| | tag_overridden | BOOLEAN |
| | evaluated_at | DATETIME |

**SCORE_DETAIL**

| | | |
|---|---|---|
| PK | **detail_id** | INT |
| FK | *score_id* | INT |
| FK | *requirement_id* | INT |
| | is_met | BOOLEAN |
| | matched_value | VARCHAR(100) |
| | weight_awarded | DECIMAL(5,2) |

## Interviews & Feedback

**AI_INSIGHT**

| | | |
|---|---|---|
| PK | **insight_id** | INT |
| FK | *application_id* | INT |
| | strengths | TEXT |
| | concerns | TEXT |
| | summary | TEXT |
| | status | ENUM(PENDING, READY) |
| | generated_at | DATETIME |

**INTERVIEW_FEEDBACK**

| | | |
|---|---|---|
| PK | **feedback_id** | INT |
| FK | *interview_id* | INT |
| FK | *author_id* | INT |
| | notes | TEXT |
| | created_at | DATETIME |

**INTERVIEW_PANEL**

| | | |
|---|---|---|
| PK,FK | **interview_id** | INT |
| PK,FK | **interviewer_id** | INT |
| | assigned_at | DATETIME |

**INTERVIEW**

| | | |
|---|---|---|
| PK | **interview_id** | INT |
| FK | *application_id* | INT |
| FK,UK | *slot_id* | INT |
| FK | *scheduled_by* | INT |
| | status | ENUM(SCHEDULED, CONDUCTED, NO_SHOW, POSTPONED, CANCELLED) |
| | google_event_id | VARCHAR(150) |
| | created_at | DATETIME |

**INTERVIEW_SLOT**

| | | |
|---|---|---|
| PK | **slot_id** | INT |
| FK | *vacancy_id* | INT |
| | slot_date | DATE |
| | start_time | TIME |
| | end_time | TIME |
| | venue | VARCHAR(150) |
| | is_booked | BOOLEAN |

**STRUCTURED_RATING**

| | | |
|---|---|---|
| PK | **rating_id** | INT |
| FK | *interview_id* | INT |
| FK | *interviewer_id* | INT |
| FK | *criterion_id* | INT |
| | rating | TINYINT (1-5) |
| | comment | VARCHAR(255) |
| | submitted_at | DATETIME |

**FEEDBACK_CRITERION**

| | | |
|---|---|---|
| PK | **criterion_id** | INT |
| UK | name | VARCHAR(80) |
| | description | VARCHAR(255) |

## Vacancies & Application Forms

**DEPARTMENT**

| | | |
|---|---|---|
| PK | **department_id** | INT |
| UK | name | VARCHAR(100) |
| | description | VARCHAR(255) |

**VACANCY**

| | | |
|---|---|---|
| PK | **vacancy_id** | INT |
| FK | *department_id* | INT |
| FK | *created_by* | INT |
| | title | VARCHAR(150) |
| | description | TEXT |
| | location | VARCHAR(100) |
| | status | ENUM(DRAFT, OPEN, CLOSED, FILLED) |
| | published_at | DATETIME |
| | closed_at | DATETIME |

**INTERVIEW_QUESTION**

| | | |
|---|---|---|
| PK | **question_id** | INT |
| FK | *vacancy_id* | INT |
| FK | *approved_by* | INT |
| | question_text | TEXT |
| | category | ENUM(TECHNICAL, BEHAVIOURAL) |
| | is_approved | BOOLEAN |
| | created_at | DATETIME |

**FORM_FIELD**

| | | |
|---|---|---|
| PK | **field_id** | INT |
| FK | *vacancy_id* | INT |
| FK | *requirement_id* | INT |
| | label | VARCHAR(150) |
| | field_type | ENUM(TEXT, NUMBER, DATE, LIST, URL) |
| | is_required | BOOLEAN |
| | display_order | INT |

**VACANCY_REQUIREMENT**

| | | |
|---|---|---|
| PK | **requirement_id** | INT |
| FK | *vacancy_id* | INT |
| | req_type | ENUM(SKILL, EXPERIENCE, QUALIFICATION, CERTIFICATION) |
| | name | VARCHAR(100) |
| | min_value | VARCHAR(50) |
| | is_mandatory | BOOLEAN |
| | weightage | DECIMAL(5,2) |

**JOB_POSTING**

| | | |
|---|---|---|
| PK | **posting_id** | INT |
| FK | *vacancy_id* | INT |
| | job_board | ENUM(LINKEDIN, INDEED) |
| | external_posting_id | VARCHAR(100) |
| | status | ENUM(POSTED, FAILED) |
| | posted_at | DATETIME |